FedCLR: Federated Contrastive Learning for Cross-Domain Recommendation

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***Abstract*—Conventional cross-domain recommendation (CDR) models require centrally collecting user data, which conflicts with user privacy expectations. Federated learning (FL) addresses this by training on local devices and aggregating model parameters, avoiding centralized data collection. However, deviations between global and local models—caused by non-independent and identically distributed (non-IID) user data—challenge existing FL-based CDR models in alleviating data sparsity and cold-start problems. This paper presents an implementation and evaluation of Fed-CLR, a novel end-to-end federated contrastive learning-based model for cross-domain recommendation. Fed-CLR uses a variational inference model to characterize interaction distributions in the source domain, reconstructs historical interactions in the target domain via a generative model, and performs federated contrastive learning at the model level (inner-model and inter-model) to reduce global–local deviations. A constraint mechanism (Con-Mec) enforces consistency between the global and local models. We reproduce and evaluate the model on three Douban cross-domain datasets (Movie&Music, Movie&Book, Music&Book) under the cold-start setting, with systematic hyperparameter tuning using Optuna grid search. Our implementation achieves competitive results consistent with the original paper.**

***Index Terms*—Federated learning, contrastive learning, variational inference, cross-domain recommendation, cold-start recommendation.**

I. Introduction

Recommender systems are central to modern digital platforms, helping users discover relevant content from a vast item catalog [1]. Collaborative filtering (CF) is one of the most widely applied paradigms, with recent advances leveraging graph neural networks (e.g., NGCF [2], LightGCN [3]) to model high-order user–item interactions. However, single-domain RSs suffer from two fundamental challenges: *data sparsity*, where historical interactions are insufficient for learning accurate preferences, and the *cold-start problem*, where users entering a new domain have no interaction history.

Cross-Domain Recommendation (CDR) addresses these limitations by transferring knowledge from a data-rich *source domain* to a data-sparse *target domain* [4]. Representative CDR methods such as EMCDR [4] learn explicit cross-domain mappings using multi-layer perceptrons. More recent VAE-based methods (e.g., SA-VAE [7]) use variational inference for flexible distribution-level transfer. Yet all of these approaches require centralizing raw user data, raising serious privacy concerns.

Federated Learning (FL) [8] offers a privacy-preserving alternative: participants train models locally and share only model parameters with a central server. FedCDR [12] introduced FL into CDR to protect user privacy, but it inherits the non-IID challenge. When user data distributions differ across clients, local gradients diverge, causing the aggregated global model to deviate from any individual local model, degrading recommendation quality and slowing convergence.

To address this, Wang et al. [17] proposed Fed-CLR, which incorporates federated contrastive learning to reduce deviations between global and local models. The key idea is to use a constraint mechanism (Con-Mec) that simultaneously maximizes the consistency between local and global models (inter-model) and minimizes similarity between consecutive local epochs (inner-model), enforcing stable and directed learning dynamics.

In this paper, we reproduce and evaluate Fed-CLR on three Douban cross-domain datasets under the cold-start setting. We implement the server-side federated aggregation and the client-side local training with variational inference and contrastive loss, and conduct systematic hyperparameter tuning using Optuna's grid search with an early-stopping (pruning) strategy. Our contributions are:

- A faithful implementation of Fed-CLR following the original algorithmic specification.
- Systematic hyperparameter tuning using Optuna grid search across learning rate and dropout, with pruning based on median Recall@50.
- Evaluation on Movie&Music, Movie&Book, and Music&Book datasets, with results averaged over five random seeds.

II. Related Work

A. Cross-Domain Recommendation

CDR methods can be broadly grouped into embedding-based and generative approaches. EMCDR [4] learns domainspecific embeddings via matrix factorization and trains a mapping function (linear or MLP) for knowledge transfer. CMF [5] performs collective matrix factorization across domains simultaneously. Both approaches use centralized training and suffer on sparse datasets. sharedVAE [6] adopts a VAE backbone but still follows EMCDR's step-wise training philosophy. SA-VAE [7] uses source-domain variational distributions as priors for the target domain, enabling end-to-end training, but remains centralized.

B. Federated Learning for Recommendation

FCF [9] is the earliest federated CF system, keeping user embeddings local. FedMF [10] enhances privacy by encrypting local gradients before upload. For CDR, FedCT [11] reformulates cross-domain transfer in a mobile edge environment, while FedCDR [12] achieves privacy-preserving CDR via FL. However, both assume near-IID data or do not explicitly address the non-IID deviation problem.

C. Contrastive Learning

Contrastive learning [13] trains models to bring similar (positive) pairs close and push dissimilar (negative) pairs apart in representation space. In recommendations, CLCRec [14] uses contrastive learning for cold-start users, and HGCL [15] applies it to heterogeneous graphs. In FL, MOON [16] pioneered model-level contrastive learning to correct local model drift. $Fed\text{-}CLR$ extends this idea to the CDR setting with domain-transfer objectives.

III. Dataset Statistics

We evaluate $Fed\text{-}CLR$ on three Douban and two Amazon cross-domain dataset pairs. This section reports the raw dataset statistics from the original paper alongside those of our downloaded datasets, highlighting any discrepancies.

A. Douban Datasets

Table I reproduces the per-domain statistics reported in the original paper [17]. Table II compares the cross-domain pair statistics—overlapping users, source items, and target items—between the paper and our downloaded version, together with relative differences (Δ).

TABLE I
Per-Domain Statistics of the Douban Dataset (from Original Paper [17])

Dataset	Domain	#Users	#Items	#Interact.	Sparsity
Douban	Book	2,718	6,778	96,041	0.9948
	Music	2,718	5,568	69,709	0.9954
	Movie	2,718	9,566	1,133,420	0.9564

The Douban datasets are nearly identical between the paper and our downloaded version. The minor discrepancies of -1 item in source and target item counts across all three domain pairs are attributable to item-id deduplication during preprocessing and have a negligible effect on results.

B. Amazon Datasets

Table III reproduces the per-domain Amazon statistics from the original paper [17]. Table **??** compares the cross-domain pair statistics for Game→Video and Cloth→Sport.

IV. Methodology

A. Problem Formulation

Let $\mathcal{U} = \{u_1, u_2, \ldots, u_N\}$ denote N overlapping participants, where each participant u_i has interactions in both the source domain $\mathcal{D}^S$ and the target domain $\mathcal{D}^T$. Let $\mathbf{x}_i^S \in \mathbb{R}^{1\times M^S}$ and $\mathbf{x}_i^T \in \mathbb{R}^{1\times M^T}$ be binary interaction vectors in the source and target domains, respectively, where M^S and M^T are the numbers of source and target domain items. The goal is to learn a global federated model M_G that predicts missing interactions in the target domain for cold-start users, without centralizing raw user data.

B. Overview of $Fed\text{-}CLR$

$Fed\text{-}CLR$ follows a federated training paradigm with a central server and N client participants. Training alternates between:

1) **Server side:** initializes and maintains the global model $M_G = (Enc_\phi, Dec_\theta)$; distributes parameters to selected clients; aggregates returned local parameters.
2) **Client side:** each selected participant u_i receives global parameters, performs local training with variational inference, reconstruction, and contrastive objectives, then returns updated parameters.

The client-side local model consists of three modules: (i) an encoder characterizing interaction distributions in the source domain; (ii) a decoder reconstructing target domain interactions; and (iii) a contrastive module ($Con\text{-}Mec$) enforcing consistency between global and local representations.

C. Server-Side Federated Aggregation

Algorithm 1 details the server-side procedure. At each global iteration m, the server randomly selects L participants, distributes the current global parameters $(\phi^{(m)}, \theta^{(m)})$, waits for locally updated parameters, and aggregates via weighted averaging:

$$\phi^{(m+1)} \leftarrow \sum_{i=1}^{L} \frac{|D_i|}{|D|}\phi_i^{(m)}, \quad \theta^{(m+1)} \leftarrow \sum_{i=1}^{L} \frac{|D_i|}{|D|}\theta_i^{(m)}, \quad (1)$$

where $|D_i|$ is the size of participant u_i's local dataset and $|D| = \sum_{i=1}^{L} |D_i|$.

D. Client-Side Local Training

Algorithm 2 details the local training procedure for participant u_i.

Module 1 — Source Domain Encoder. For participant u_i with source interactions $\mathbf{x}_i^S$, we model the latent behavior distribution using variational inference. We approximate the intractable posterior $p_\theta(\mathbf{z}_i|\mathbf{x}_i^S)$ with a Gaussian variational distribution:

TABLE II
DOUBAN CROSS-DOMAIN PAIR STATISTICS: PAPER VS. OURS. Δ DENOTES RELATIVE DIFFERENCE (%) OF OUR DATASET W.R.T. THE ORIGINAL PAPER.

Domain Pair	**Paper Dataset**			**Our Dataset**			**Δ (%)**		
	Overlap	**Src Items**	**Tgt Items**	**Overlap**	**Src Items**	**Tgt Items**	**Overlap**	**Src Items**	**Tgt Items**
Movie→Music	1,666	9,566	5,568	1,666	9,555	5,567	0.0	−0.1	−0.0
Movie→Book	2,106	9,566	6,778	2,106	9,555	6,777	0.0	−0.1	−0.0
Music→Book	1,566	5,568	6,778	1,566	5,567	6,777	0.0	−0.0	−0.0

TABLE III
PER-DOMAIN STATISTICS OF THE AMAZON DATASET (FROM ORIGINAL PAPER [17])

Dataset	**Domain**	**#Users**	**#Items**	**#Interact.**	**Sparsity**
Amazon	Sport	27,328	12,655	170,426	0.9995
	Cloth	41,829	17,943	194,121	0.9997
	Game	25,025	12,319	157,721	0.9995
	Video	19,457	8,751	158,961	0.9991

Algorithm 1 The Server Side of $Fed\text{-}CLR$: The mth ($m \geq 0$) Global Iteration.

Input: upper limit of global iterations NGI.
Output: global federated cross-domain recommendation model M_G.
Initialize model parameters $\phi^{(m)}$ and $\theta^{(m)}$ of M_G, including encoder Enc_ϕ and decoder Dec_θ;
repeat
    $U^L \leftarrow$ randomly select L participants from client side;
    for each selected participant $u_i \in U^L$ **in parallel do**
        Distribute global model parameters $\phi^{(m)}$ and $\theta^{(m)}$ to u_i;
        $\phi_i^{(m)}, \theta_i^{(m)} \leftarrow \text{LocalTraining}(u_i, \phi^{(m)}, \theta^{(m)})$;
    end for
    $\phi^{(m+1)} \leftarrow \sum_{i=1}^{L} \frac{|D_i|}{|D|} \phi_i^{(m)}$;
    $\theta^{(m+1)} \leftarrow \sum_{i=1}^{L} \frac{|D_i|}{|D|} \theta_i^{(m)}$;
    $M_G \leftarrow (\phi^{(m+1)}, \theta^{(m+1)})$;
    $m{+}{+}$;
until M_G converges or $m \geq NGI$
return M_G;

$$q_\phi^{(e)}\left(z_i^{(e)} \mid \mathbf{x}_i^S\right) = \mathcal{N}\left(z_i^{(e)} \mid \mu_\phi^{(e)}(\mathbf{x}_i^S),\, \text{diag}\left[(\sigma_\phi^{(e)})^2(\mathbf{x}_i^S)\right]\right). \tag{2}$$

To enable backpropagation through the sampling step, the reparameterization trick is applied:

$$z_i^{(e)} = \mu_i^{(e)} + \epsilon \odot \sigma_i^{(e)}, \quad \epsilon \sim \mathcal{N}(0, I). \tag{3}$$

At the first local epoch ($e = 0$), $z_i^{(0)}$ is set as z_{glob}, the encoding of the global model, to initialize the contrastive baseline.

Module 2 — Target Domain Decoder. The decoder $p_\theta^{(e)}(\mathbf{x}_i^T | z_i^{(e)})$ reconstructs target domain interactions from the latent representation $z_i^{(e)}$, using a mirror feedforward network with Tanh activations. The reconstruction objective combines a softmax-based likelihood and a KL regularization term (ELBO):

$$\begin{aligned}\mathcal{L}_{\text{rec}} = \mathbb{E}_q&\left[\log p_\theta^{(e)}\left(\mathbf{x}_i^T \mid z_i^{(e)}\right)\right] \\ &- \text{KL}\left[q_\phi^{(e)}\left(z_i^{(e)} \mid \mathbf{x}_i^S\right) \,\|\, p_\theta^{(e)}\left(z_i^{(e)}\right)\right].\end{aligned} \tag{4}$$

The KL term has the closed-form solution:

$$-\text{KL} = \tfrac{1}{2}\left(\log(\sigma_i^{(e)})^2 - (\mu_i^{(e)})^2 - (\sigma_i^{(e)})^2 + 1\right). \tag{5}$$

Module 3 — Federated Contrastive Learning ($Con\text{-}Mec$). This module computes the contrastive loss using three representation vectors: the current epoch's latent vector $z_i^{(e)}$, the previous epoch's latent vector $z_i^{(e-1)}$, and the global model's behavior representation z_{glob}. All vectors are first ℓ_2-normalized:

$$\tilde{z} = \frac{z}{\|z\|_2}. \tag{6}$$

The inner-product similarities are computed as:

$$\text{SIM(InnM)} = \tilde{z}_i^{(e)} \cdot \tilde{z}_i^{(e-1)}, \tag{7}$$

$$\text{SIM(IntM)} = \tilde{z}_i^{(e)} \cdot \tilde{z}_{\text{glob}}. \tag{8}$$

The contrastive loss is:

$$\mathcal{L}_{\text{con}} = -\log \frac{\exp(\text{SIM(IntM)}/\tau)}{\exp(\text{SIM(IntM)}/\tau) + \exp(\text{SIM(InnM)}/\tau)}, \tag{9}$$

where τ is the temperature coefficient. The inter-model term (IntM) treats the global model as the positive anchor (encouraging the local model to stay aligned with the global), while the inner-model term (InnM) treats the previous-epoch representation as the negative (penalizing stagnation and encouraging diverse update directions). The overall local loss is:

$$\mathcal{L}_{\text{local}} = \mathcal{L}_{\text{rec}} + \alpha \cdot \mathcal{L}_{\text{con}}, \tag{10}$$

where α controls the weight of the contrastive objective. In the first local epoch ($e = 0$), $\mathcal{L}_{\text{local}} = \mathcal{L}_{\text{rec}}$ since no previous-epoch representation is available.

Algorithm 2 The Client Side of $Fed\text{-}CLR$: LocalTraining(u_i, ϕ, θ).

Input: training data $\mathbf{x}_i^S$, $\mathbf{x}_i^T$; upper limit of local epochs E; temperature τ; contrastive weight α; learning rate η.
Output: parameters $\phi_i^{(e)}$, $\theta_i^{(e)}$ of local model M_i.
$e \leftarrow 0; \quad (Enc_{\phi_i^{(e)}}, Dec_{\theta_i^{(e)}}) \leftarrow (\phi, \theta)$;
repeat
  $z_i^{(e)} \leftarrow Enc_{\phi_i^{(e)}}(\mathbf{x}_i^S)$;
  if $e == 0$ **then**
    $z_{\text{glob}} \leftarrow z_i^{(e)}$;
  end if
  $\hat{\mathbf{x}}_i^T \leftarrow Dec_{\theta_i^{(e)}}(\mathbf{x}_i^S)$;
  $\mathcal{L}_{\text{rec}} \leftarrow \text{CalRecLoss}(\hat{\mathbf{x}}_i^T, \mathbf{x}_i^T)$;
  if $e \geq 1$ **then**
    $z_i^{(e-1)} \leftarrow Enc_{\phi_i^{(e-1)}}(\mathbf{x}_i^S)$;
    $\text{SIM(InnM)} \leftarrow z_i^{(e)} \cdot z_i^{(e-1)}$;
    $\text{SIM(IntM)} \leftarrow z_i^{(e)} \cdot z_{\text{glob}}$;
    $\mathcal{L}_{\text{con}} \leftarrow \text{CalContraLoss}(\text{SIM(InnM)}, \text{SIM(IntM)})$;
    $\mathcal{L}_{\text{local}} \leftarrow \mathcal{L}_{\text{rec}} + \alpha \cdot \mathcal{L}_{\text{con}}$;
  else
    $\mathcal{L}_{\text{local}} \leftarrow \mathcal{L}_{\text{rec}}$;
  end if
  $\phi_i^{(e+1)} \leftarrow \phi_i^{(e)} - \eta \frac{\partial \mathcal{L}_{\text{local}}}{\partial \phi_i^{(e)}}$;
  $\theta_i^{(e+1)} \leftarrow \theta_i^{(e)} - \eta \frac{\partial \mathcal{L}_{\text{local}}}{\partial \theta_i^{(e)}}$;
  $M_i \leftarrow (\phi_i^{(e+1)}, \theta_i^{(e+1)})$;
  $e{+}{+}$;
until M_i converges or $e \geq E$
return $\phi_i^{(e)}, \theta_i^{(e)}$;

V. Experimental Setup

A. Datasets

We evaluate $Fed\text{-}CLR$ on three Douban cross-domain datasets: Movie&Music, Movie&Book, and Music&Book [17]. In each pair, the sparser domain is treated as the target domain. We use the standard cold-start split: 20% of users are randomly designated as cold-start users (testing set), and the remaining 80% are used for training. Reported results are averages over five random seeds (42–46) to ensure statistical reliability. Detailed dataset statistics and a comparison with the original paper's datasets are provided in Section III.

We additionally evaluate $Fed\text{-}CLR$ on two Amazon cross-domain dataset pairs: Game→Video and Cloth→Sport. Due to the large scale of these datasets, 10% of overlapping users are randomly sampled before performing the 80/20 train–cold-start split. Amazon results are reported for a single seed (42).

B. Hardware and Software Configuration

Experiments were run on the following hardware and software environment:

- **CPU:** 13th Gen Intel Core i7-13620H (10 cores, 16 logical processors) (2.40 GHz)
- **RAM:** 16.00 GB DDR5
- **OS:** Microsoft Windows 11 Home Single Language
- **Python:** 3.13.9
- **PyTorch:** 2.10.0+cu130
- **CUDA:** 13.0
- **Optuna:** 4.8.0

C. Fixed Hyperparameters

The following hyperparameters were kept constant across all datasets, informed by convergence graphs and ablation experiments:

- **Communication rounds:** 50 (determined by visual inspection of convergence curves; consistent with the original paper's training graphs)
- **Local epochs per round:** 5 (experiments with 10 local epochs degraded performance)
- **Fraction of clients sampled per round (C):** 0.1. The original paper does not explicitly specify the number of clients sampled per communication round; we therefore adopt the standard federated learning convention of sampling a fraction $C = 0.1$ of all available users per round.
- **Temperature coefficient τ:** 0.3 (as reported optimal in the original paper)
- **Contrastive weight α:** 40 (as reported optimal in the original paper)
- **VAE architecture:** $M^S \to 300 \to 100 \to 300 \to M^T$
- **Optimizer:** Adam
- **Parameter initialization:** Xavier Glorat
- **Mini-batch size:** 128

D. Hyperparameter Tuning

Two hyperparameters were tuned using Optuna grid search: **learning rate** and **dropout rate**. These parameters are not explicitly reported in the original paper and thus required exploration.

- **Learning rate search range:** $\{0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5\}$ (8 values, as in the original paper)
- **Dropout search range:** $[0.0, 1.0]$ with step size 0.1 (11 values, as in the original paper)

Tuning was conducted with the following strategy: each trial ran for 10 communication rounds; the first five trials were never pruned; from the sixth trial onward, after every two rounds of training, Recall@50 was evaluated and compared to the median Recall@50 of all completed trials—if the current trial fell below the median, it was pruned. This Optuna median pruner strategy significantly reduced the total search time while retaining promising configurations.

The best hyperparameter configurations found for each dataset are summarized in Table IV.

VI. Results

A. Baseline Models

Table V summarises the baseline models considered in this study alongside their implementation status. All baseline figures cited in the results are taken directly from the original $Fed\text{-}CLR$ paper [17].

TABLE IV
BEST TUNED HYPERPARAMETERS PER DATASET

Dataset	Learning Rate	Dropout
Movie&Music	0.05	0.6
Movie&Book	0.01	0.0
Music&Book	0.01	0.7
Game→Video (dropout=0.7)	0.1	0.7
Game→Video (dropout=0.5)	0.1	0.5
Cloth→Sport	0.05	0.4

TABLE V
BASELINE MODELS AND IMPLEMENTATION STATUS

Model	Type	Status
SA-VAE [7]	Non-federated CDR (VAE-based)	In progress
CMF [5]	Non-federated CDR (matrix factorization)	Pending
FedCDR [12]	Federated CDR	Pending
sharedVAE [6]	Non-federated CDR (VAE-based)	Pending

B. Performance on Cold-Start Recommendation (Douban)

Table VI reports the performance of our $Fed\text{-}CLR$ implementation on all three Douban datasets under the 20% cold-start setting, averaged over five random seeds (42–46). We report Recall, NDCG, and Precision at both top-50 and top-100. The Δ columns indicate the relative percentage difference between our implementation and the original paper's reported $Fed\text{-}CLR$ results, computed as $\Delta = \frac{\text{Ours}-\text{Paper}}{\text{Paper}} \times 100\%$.

The results demonstrate that $Fed\text{-}CLR$ achieves consistent performance across all three cross-domain scenarios. For Movie&Music and Movie&Book our implementation falls below the paper's reported numbers by roughly 19–31%. Several implementation gaps contribute to this discrepancy: (i) the original paper does not specify the exact number of clients sampled per communication round, so we adopted the standard FL convention of $C = 0.1$; (ii) the exact number of local epochs per round is not reported, and our choice of 5 local epochs may differ from the authors' setting; and (iii) the optimal learning rate and dropout were not disclosed, requiring independent tuning from the search ranges provided in the paper, which may not have converged to the same configuration used by the authors. Notably, for Music&Book our implementation matches or surpasses the paper's figures, with NDCG improvements of up to $+41.1\%$, suggesting that the hyperparameter search recovered a particularly strong configuration for that domain pair.

C. Performance on Cold-Start Recommendation (Amazon)

Table VII reports the performance of $Fed\text{-}CLR$ on the two Amazon domain pairs. Due to dataset size, results use 10% sampled users and a single seed (42). Two dropout configurations are evaluated for Game→Video.

The Amazon datasets present a considerably harder recommendation scenario than Douban, as reflected by the substantially lower metric values. The Cloth→Sport pair yields stronger performance than Game→Video across all metrics, consistent with the relatively higher interaction density in the clothing/sports categories.

VII. DISCUSSION

Hyperparameter sensitivity. Dropout was found to have a significant effect: Movie&Book performed best with no dropout ($p = 0.0$), while Movie&Music and Music&Book benefited from moderate-to-high dropout (0.6 and 0.7, respectively). This suggests domain-pair-specific regularization requirements. The learning rate of 0.05 for Movie&Music versus 0.01 for the book-containing pairs also reflects different optimization landscapes.

Local epoch choice. Increasing local epochs from 5 to 10 was found to degrade performance, consistent with prior FL literature showing that more local updates increase client drift under non-IID data. Keeping local epochs at 5 strikes a balance between local convergence and global consistency.

Communication rounds. Visual inspection of training curves suggested convergence around 50 rounds for all datasets, consistent with the $Fed\text{-}CLR$ paper's reported convergence behavior. The Con-Mec constraint contributes to this fast convergence by aligning local update directions with the global model.

Client sampling fraction. The original paper does not specify the number or fraction of clients sampled per communication round. We adopt $C = 0.1$ following the standard federated learning convention. For the large-scale Amazon datasets, the same fraction is applied after a 10% user subsample, yielding manageable round sizes while preserving the federated training protocol.

Reproducibility. Averaging over five seeds (42–46) provides reliable estimates of model performance on the Douban datasets. Variance across seeds was low in all cases, indicating stable training dynamics.

VIII. CONCLUSION

This paper presents an implementation and evaluation of $Fed\text{-}CLR$, a federated contrastive learning model for cross-domain recommendation. We reproduced the full federated training pipeline—including variational inference, target domain reconstruction, and the $Con\text{-}Mec$ contrastive constraint—and conducted systematic hyperparameter tuning using Optuna. Our results on the Douban Movie, Book, and Music datasets confirm the effectiveness of the approach under the cold-start setting, with competitive or superior performance on Music&Book. Preliminary results on Amazon Game→Video and Cloth→Sport demonstrate feasibility on large-scale datasets. Future work will explore the model's performance with varying numbers of clients, additional Amazon domain pairs, and integration of differential privacy mechanisms. A full re-implementation of the SA-VAE baseline is also planned for direct federated vs. non-federated comparison.

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TABLE VI

COLD-START RECOMMENDATION PERFORMANCE OF *Fed-CLR* ON DOUBAN DATASETS (OURS VS. PAPER) – AVERAGED OVER SEEDS 42–46. Δ DENOTES RELATIVE DIFFERENCE (%) W.R.T. THE ORIGINAL PAPER.

Dataset	**Source**	**NDCG@50**	**NDCG@100**	**Prec@50**	**Prec@100**	**Recall@50**	**Recall@100**
Movie&Music	Paper (Fed-CLR)	0.0948	0.1034	0.0602	0.0497	0.1018	0.1538
	Ours	0.07254	0.08080	0.04168	0.03388	0.08242	0.12436
	Δ (%)	-23.5	-21.9	-30.8	-31.8	-19.0	-19.1
Movie&Book	Paper (Fed-CLR)	0.1472	0.1549	0.0817	0.0613	0.1416	0.1903
	Ours	0.11370	0.12228	0.06304	0.04802	0.10576	0.15466
	Δ (%)	-22.8	-21.1	-22.8	-21.7	-25.3	-18.7
Music&Book	Paper (Fed-CLR)	0.0848	0.0948	0.0521	0.0425	0.0993	0.1523
	Ours	0.11968	0.12812	0.06994	0.05400	0.10176	0.15108
	Δ (%)	$+41.1$	$+35.1$	$+34.2$	$+27.1$	$+2.5$	-0.8

TABLE VII

COLD-START RECOMMENDATION PERFORMANCE OF *Fed-CLR* ON AMAZON DATASETS (SEED 42, 10% USER SAMPLE).

Dataset	**Config**	**NDCG@50**	**NDCG@100**	**Prec@50**	**Prec@100**	**Recall@50**	**Recall@100**
Game$\rightarrow$Video	LR=0.1, Dropout=0.7	0.0094	0.0122	0.0019	0.0017	0.0148	0.0236
	LR=0.1, Dropout=0.5	0.0081	0.0113	0.0024	0.0020	0.0133	0.0261
Cloth$\rightarrow$Sport	LR=0.05, Dropout=0.4	0.0233	0.0296	0.0051	0.0038	0.0369	0.0579

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